Sumaya Jahan Othye

Software Engineer | AI/ML Enthusiast

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Profile

Passionate and driven Computer Science & Engineering graduate from the University of Asia Pacific (Graduated: 2026) with a strong academic foundation (CGPA 3.48). I specialize in Full-Stack Web Development and AI/ML (Deep Learning and Computer Vision), combining analytical thinking with technical problem-solving to build modern, scalable, and user-friendly digital solutions. Through my undergraduate thesis on Deepfake Image Detection using TensorFlow, I gained hands-on expertise in implementing self-supervised learning and computer vision models. I am highly motivated to innovate, collaborate, and create intelligent software applications that solve real-world problems.

Skills

- **Programming Languages:** C++, Python, JavaScript, C, Java
- **Frontend Technologies:** React.js, HTML5, CSS3, Tailwind CSS, Bootstrap
- **Backend Frameworks:** Django (Python)
- **AI/ML & Data Science:** TensorFlow, OpenCV, NumPy, Pandas, Scikit-Learn, Matplotlib
- **AI/ML Concepts:** Deep Learning, Computer Vision, CNN/ResNet Architectures, Self-Supervised Learning, Contrastive Learning, Attention Mechanisms, Model Evaluation
- **Database Management:** MySQL, SQLite, Relational Schema Design
- **Core CS Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems
- **Testing & Automation:** Selenium (Python), Automated Web Testing
- **Developer Tools:** Git, GitHub, VS Code, Figma
- **Professional Skills:** Project Management, Task Planning, Team Collaboration, Ubuntu/Windows OS

Thesis & Research

Deepfake Image Detection Using Self-Supervised Contrastive Learning and Slot Attention

University of Asia Pacific | Undergraduate Thesis | June 2026

- **GitHub:** Repository Link
- **Contributions:** Designed and implemented a custom ResNet encoder and an 8-slot Slot Attention module in TensorFlow, incorporating self-supervised contrastive learning.
- **Approach:** Combined contrastive representation learning with Slot Attention to decompose image features into discrete slots, helping the model focus on manipulated facial regions rather than global image statistics.
- **Tools:** TensorFlow, OpenCV, NumPy, Matplotlib for model development, preprocessing, and result visualization.
- **Results:** Achieved a **90.73% test accuracy** on the OpenDeepfake Preview dataset (19,999 images) by localizing facial tampering features and reducing background texture overfitting.

Projects

Academic Project Management & Collaboration Platform (Full-Stack)

Tech Stack: Python, Django, SQLite, Bootstrap, JavaScript

- Developed a web application to facilitate team collaboration, task planning, and project submissions.
- Implemented secure role-based authentication, user dashboards, and interactive progress-tracking elements.
- Designed relational database schemas in SQLite to handle concurrent data and team assignments efficiently.
- Built the backend using Django's MVT architecture and ORM, with server-side views, models, and form validation.
- Developed a responsive frontend using Bootstrap and JavaScript for dynamic, interactive dashboards.

Education

B.Sc. in Computer Science and Engineering

University of Asia Pacific

- **Duration:** 2022 – 2026
- **Result:** CGPA 3.48 (out of 4.00)
- **Status:** Graduated (2026)

Languages & Interests

- **Languages:** English, Bangla
- **Interests:** Artificial Intelligence, Machine Learning, Computer Vision, Web Development, Academic Research, Quantitative Analysis